

MATTHEW R. SIEBERT, Ph.D. - CV

114 S Highland Ave, Baltimore, MD, 21224

603-321-5469 ◊ msiebert@stsci.edu ◊ <https://msiebert1.github.io/>

RESEARCH INTERESTS

- Connecting Type Ia supernova (SN Ia) physics and cosmology via spectroscopy.
- Understanding the diversity within the white dwarf supernova zoo.
- Discovering the most distant explosions and constraining the evolution in the properties of transients with cosmic time.

APPOINTMENTS

Affiliation: Space Telescope Science Institute

STScI Postdoctoral Fellowship - STScI

2022-present

(50% independent science + 50% support of the HST STIS instrument team)

- Flux Recalibration Lead: Led a team to develop and release new calibration files across the STIS first-order and echelle modes, following updates to the atmospheric models of the three primary standard stars. Coordinated the fourth and fifth reference-file releases (the most complex to date) and authored two related ISRs on UV echelle calibration.
- User Support Lead: Management and training of the team of STIS Contact Scientists (CSs) and help desk fulfillers. Coordinated proposal distribution among CSs; approved program changes, coordinating with the TTRB/SPD for complex cases; oversaw the transition to reduced-gyro mode and the first “Flexible Thursday” implementations; reviewed a variety of STIS programs for detector safety.
- Science: Spectroscopic lead for the high-redshift SN effort within the Transient Science at Space Telescope group. Four first-author publications including the characterization of the first high-redshift SNe with JWST and multi-wavelength spectroscopic studies of local rare white dwarf SNe. Co-PI of an accepted Cycle 5 JWST program for high-redshift SNe Ia and led a Cycle 30 HST program to obtain STIS UV spectra of local high-velocity SNe Ia.

NSF Graduate Fellow / Graduate Research Assistant - UC Santa Cruz

2016-2022

- PhD Thesis: *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data* (Advisor: Ryan J. Foley).

Science: Four first-author publications including studies on the relationship between SN Ia spectral properties and cosmological distance measurements, and progenitor scenarios of rare white dwarf SNe.

Software: Developed an open-source SQL database and suite of analysis tools (Kaepora) for spectroscopy of SNe Ia and the UCSC spectroscopic reduction pipeline.

REU Intern - MIT Haystack Observatory

2015

- User-interface development for the Haystack 37m radio telescope

EDUCATION

University of California Santa Cruz, Santa Cruz, CA PhD: Astronomy and Astrophysics	June 2022
University of California Santa Cruz, Santa Cruz, CA MS: Astronomy and Astrophysics	December 2018
Cornell University, Ithaca, NY BS: Engineering Physics	June 2016

AWARDS

GECO Postdoctoral Fellowship (UVa)	September 2026
STScI Achievement Award - STIS Flux Recalibration Team	2024
STScI Postdoctoral Fellowship	2022
Las Cumbres Observatory Postdoctoral Fellowship - <i>Declined</i>	2022
UC Santa Cruz Dissertation Fellowship	2021
UC Santa Cruz Department of Astronomy Whitford Prize	2018
NSF Graduate Research Fellowship	2018

PUBLICATIONS

First author: 8 (191 citations) [ADS], All: 109 (8846 citations), h-index: 41 [ADS]

First Author Publications

1. *SN 2025ogs: A Spectroscopically-Normal Type Ia Supernova at $z = 2$ as a Benchmark for Redshift Evolution*
Siebert, M. R., Pierel, J. D. R.; Engesser, M.; Coulter, D. A.; DeCoursey, C.; Fox, O. D.; Rest, A. et al. (20 authors)
ApJL Volume 1002, Issue 1, id.L3, 12 pp. 2026. [arXiv:2512.19783]
2. *Discovery of a Relativistic Stripped Envelope Type Ic-BL Supernova at $z = 2.83$ with JWST*
Siebert, M. R.; C. DeCoursey; D. A. Coulter; M. Engesser; J. D. R. Pierel; A. Rest; E. Egami; (22 authors)
ApJL Volume 972, Issue 1, id.L13, 13 pp. 2024. [arXiv:2406.05076]
3. *Ground-based and JWST Observations of SN 2022pul. I. Unusual Signatures of Carbon, Oxygen, and Circumstellar Interaction in a Peculiar Type Ia Supernova*
Siebert, M. R.; Kwok, L. A.; Johansson, J., ; Jha, S. W.; Blondin, S.; Dessart, L.; Foley, R. J.; Hillier, D. J.; Larison, C.; Pakmor, R.; Temim, T. (82 authors)
ApJ, Volume 960, Issue 1, id.88, 19 pp. 2023. [arXiv:2308.12449]
4. *An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star*
Siebert, M. R.; Foley, R. J.; Zenati, Y.; Dimitriadis, G.; Schmidt, E.; Yang, G.; Davis,

- K. W.; Taggart, K.; Rojas-Bravo, C.
ApJ, Volume 958, Issue 2, id.173, 14 pp. 2023. [\[arXiv:2306.11788\]](#)
5. *Strong Calcium Emission Indicates that the Ultraviolet-flashing SN Ia 2019yvq Was the Result of a Sub-Chandrasekhar-mass Double-detonation Explosion*
Siebert, M. R.; Dimitriadis, G.; Polin, A.; and Foley, R. J.
ApJL, Volume 900, Issue 2, article id.L27, 11 pp. 2020. [\[arXiv:2007.13793\]](#)
 6. *A Possible Distance Bias for Type Ia Supernovae with Different Ejecta Velocities*
Siebert, M. R.; Foley, R. J.; Jones, D. O.; and Davis, K. W.
MNRAS, Volume 493, Issue 4, pp.5713-5725. 2020. [\[arXiv:2002.09490\]](#)
 7. *Investigating the Diversity of Type Ia Supernova Spectra with the Open-Source Relational Database Kaepora*
Siebert, M. R.; Foley, R. J.; Jones, D. O.; Angulo, R.; Davis, K. W.; Duarte, A.; Strasburger, E.; Conlon, M.; Kazmi, N.; Nishimoto, R.; Schubert, M.; Sun, L.; and Tippens, R.
MNRAS, Volume 486, Issue 4, p.5785-5808. 2019. [\[arXiv:1905.02204\]](#)
 8. *The Unprecedented Properties of the First Electromagnetic Counterpart to a Gravitational-wave Source*
Siebert, M. R.; Foley, R. J.; Drout, M. R.; Kilpatrick, C. D.; Shappee, B. J.; Coulter, D. A.; Kasen, D.; Madore, B. F.; Murguia-Berthier, A.; Pan, Y. -C.; Piro, A. L.; Prochaska, J. X.; Ramirez-Ruiz, E.; Rest, A.; Contreras, C.; Morrell, N.; Rojas-Bravo, C.; Simon, J. D.
ApJL, Volume 848, Issue 2, article id. L26, 6 pp. 2017. [\[arXiv:1710.05440\]](#)

Major Contribution Publications

9. *A spectroscopically confirmed, strongly lensed, metal-poor Type II supernova at $z = 5.13$*
Coulter, D. A.; Larison, C.; Pierel, J. D. R.; Fujimoto, S.; Kokorev, V.; Allingham, J. F. V.; Moriya, T. J.; **Siebert, M. R.** et al. (89 authors)
Submitted to Science. 2026. [\[arXiv:2601.04156\]](#)
10. *Discovery of a likely Type II SN at $z = 3.6$ with JWST*
Coulter, D. A.; Pierel, J. D. R.; DeCoursey, C.; Moriya, T. J.; **Siebert, M. R.** et al. (33 authors)
ApJ. Volume 1002, Issue 1, id.83, 11 pp. 2026. [\[arXiv:2406.05060\]](#)
11. *The New Status Quo? SN 2021qvo is Another 2003fg-like Type Ia Supernova with a Rising Light-Curve Bump*
Paniagua, A. et al. (27 authors incl. **Siebert, M. R.**)
ApJ. Volume 997, Issue 2, id.261, 16 pp.2026. [\[arXiv:2508.13263\]](#)
12. *Feeling Blue: Constructing a Robust SALT3 UV Template and Constraining its Redshift Dependency*
Wang, Q.; Jones, D. O.; Pierel, J. D. R.; **Siebert, M. R.** et al. (16 authors)
Submitted to ApJ. 2025. [\[arXiv:2512.25064\]](#)
13. *Testing for Intrinsic Type Ia Supernova Luminosity Evolution at $z > 2$ with JWST*
Pierel, J. D. R.; Coulter, D. A.; **Siebert, M. R.** et al. (41 authors)
ApJL. Volume 981, Issue 1, id.L9, 10 pp. 2025. [\[arXiv:2411.11953\]](#)
14. *Discovery of an Apparent Red, High-velocity Type Ia Supernova at $z = 2.9$ with JWST*

- Pierel, J. D. R.; Engesser, M.; Coulter, D. A.; DeCoursey, C.; **Siebert, M. R.** et al. (30 authors)
ApJ, Volume 971, Issue 2, id.L32, 11 pp. 2024. [[arXiv:2406.05089](#)]
15. *Ground-based and JWST Observations of SN 2022pul: II. Evidence from Nebular Spectroscopy for a Violent Merger in a Peculiar Type-Ia Supernova*
Kwok, L. A.; **Siebert, M. R.**; Johansson, J.; Jha, S. W.; Blondin, S.; Dessart, L.; Foley, R. J.; Hillier, D. J.; Larison, C.; Pakmor, R.; Temim, T. (82 authors)
ApJ, Volume 960, Issue 1, id.88, 19 pp. 2024. [[arXiv:2308.12450](#)]
 16. *Understanding Type Ia Supernova Distance Biases by Simulating Spectral Variations*
Pierel, J. D. R.; Jones, D. O.; Dai, M.; Adams, D. Q.; Kessler, R.; Rodney, S.; **Siebert, M. R.**; Foley, R. J.; Kenworthy, W. D.; Scolnic, D.
ApJ, Volume 911, Issue 2, id.96, 13 pp. 2021. [[arXiv:2012.07811](#)]
 17. *Three Hypervelocity White Dwarfs in Gaia DR2: Evidence for Dynamically Driven Double degenerate Double-detonation Type Ia Supernovae*
Shen, K. J. et al. (26 authors incl. **Siebert, M. R.**)
ApJ, Volume 865, Issue 1, article id. 15, 14 pp. 2018. [[arXiv:1804.11163](#)]

Technical Publications (Unrefereed)

18. *Rederivation of STIS Secondary Echelle Mode Traces*
Siebert, M. R.; Monroe, T.; Hernandez, S.
Instrument Science Report [[STIS ISR 2024-03](#)].
19. *Recalibration of Pre-SM4 STIS Echelle Throughputs*
Siebert, M. R.; Carlberg, J. K. ; Hernandez, S. ; Monroe, T.
Instrument Science Report [[STIS ISR 2024-02](#)].
20. *Safety Acquisitions: Redundancy for non-repeatable multi-orbit STIS visits*
Dallas, M. M.; **Siebert, M. R.**
Instrument Science Report [[STIS ISR 2024-01](#)].

OBSERVING PROGRAMS

PI or Co-PI

- **James Webb Space Telescope:** 2026
Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$
(121 hours, GO-12146, Co-PI)
- **Hubble Space Telescope:** 2022
Measuring the Effect of Progenitor Metallicity on Type Ia Supernova Distance Estimates
(8 orbits, GO-17170, PI)
- **Gemini South:** 2021
Host-Galaxy Spectroscopy of Type Ia Supernovae
(4 hours, GS-2021B-Q-243, PI)
- **Gemini South:** 2020
Host-Galaxy Spectroscopy of Type Ia Supernovae
(5.8 hours, GS-2020B-Q-325, PI)

Major Contributions as Co-I

- **James Webb Space Telescope:** *2026*
JWST Multi-Cycle Deep Transient Survey in GOODS-S
(43 hours, GO-12577)
- **James Webb Space Telescope:** *2025*
Constraining the Hubble Constant with Nebular-Phase Spectroscopy of a 20x Magnified, Multiply Imaged Supernova
(7 hours, GO-12503)
- **James Webb Space Telescope:** *2025*
SN Eos: A Multiply-Imaged, 30x Magnified SN Near the Epoch of Reionization
(10 hours, GO-9493)
- **James Webb Space Telescope:** *2025*
JWST Multi-Cycle Deep Transient Survey in GOODS-S
(45 hours, GO-8060)
- **James Webb Space Telescope:** *2024*
A Luminous, Red Transient at $z = 3$: An Extreme Test Bed for Supernova Evolution
(9 hours, GO-9372)
- **James Webb Space Telescope:** *2024*
Do Pass $z=2$, Do Collect Type Ia Supernovae: Breaking Out of Redshift Jail with JWST
(76 hours, GO-5324)
- **James Webb Space Telescope:** *2023*
The High- z Menagerie: A Rare Chance to Study the Early and Exotic Transient Universe
(9 hours, GO-6585)
- **James Webb Space Telescope:** *2023*
JWST NIRSpec/NIRCam Follow-Up of the High-Redshift Transients Discovered in the GOODS-S JADES-Deep Field
(18 hours, GO-6541)
- **Hubble Space Telescope:** *2023*
Flash Thursday: Early-time UV Spectroscopy of Type II Supernovae with HST
(21 orbits, GO-17807)
- **Hubble Space Telescope:** *2022*
Going Up on a Thursday: Flash Spectroscopy of Type II Supernovae with HST
(21 orbits, GO-17507)
- **Hubble Space Telescope:** *2021*
Measuring the Effect of Progenitor Metallicity on Type Ia Supernova Distance Estimates
(12 orbits, GO-16690)
- **Hubble Space Telescope:** *2020*
Measuring the Effect of Progenitor Metallicity on Type Ia Supernova Distance Estimates
(12 orbits, GO-16238)

SOFTWARE

Kaepora - An open-source relational database and suite of analysis tools for the study of SN Ia spectra (<https://github.com/msiebert1/kaepora>).

The UCSC Transients Spectral Reduction Pipeline - A reduction pipeline developed with supernova observations in mind for spectroscopic data from the LRIS and Kast spectrographs at Keck Observatory and Lick Observatory, respectively (https://github.com/msiebert1/UCSC_spectral_pipeline).

SERVICE, TEACHING, AND OUTREACH

STScI Service

JWST Summer School: High Redshift Transients with JWST Instructor	2025
STIS User Support Lead	2024-present
STIS Flux Recalibration Working Group Lead	2023-present
HST Cycle 32 SPG Manager	2024
HST Cycle 31 TAC Leveler	2023

Mentorship

Arnav Swaroop: <i>Spectroscopic Modeling of Type Ia Supernova Light Echos (Kaepora)</i> (UC Berkeley)	2025
Xander Jenkin: <i>A Comprehensive SED Library of Type Ia Supernovae for the STIS ETC</i> (Kaepora) (STScI SASP mentee)	2024
Matilde Mingozi: <i>The Impact of STIS CCD Rotation on the E1 Wavelength Solution</i> (STIS Training Project Mentee)	2024
Eva Schmidt: <i>Strange Double-Peaked Calcium Emission in the Type Ia SN 2020hvf</i> (Ron Ruby Scholarship)	2021-2022
Erika Strasburger: <i>Constraining the Asymmetry of Type Ia Supernovae (Kaepora)</i> (NSF Graduate Research Fellowship, Dean's Thesis Award (UCSC), attended UC Berkeley graduate program)	2019-2021
Rodrigo Angulo: <i>Unburned carbon in the Outer Layers of Type Ia Supernova Ejecta (Kaepora)</i> (Davidsen Fellowship (JHU), Dean's Thesis Award (UCSC), attending JHU graduate program)	2018-2021
Kyle Davis: <i>A Comprehensive Study of SN Ia Ejecta Velocities (Kaepora)</i> (attending UCSC graduate program, postdoc at CfA-Fall 2026, active Kaepora developer)	2018-2021
Audrey Do: <i>Association of SN Ia Spectra with their Detailed Light-Curve Properties</i> (Kaepora)	2018
Rey Cervantes: <i>Consistent Characterization of SN Ia Spectral Features (Kaepora)</i>	2018
Andres Duarte: <i>New Techniques for the Homogenization of SN Ia Spectral Data</i> (Kaepora)	2017-2018

Teaching Assistant

Introduction to Research in Physics and Astrophysics (ASTR9)	2018
Exploring the Universe with Astronomical Data (ASTR8)	2018
Physics of Stars (ASTR112)	2017

Outreach

Space Astronomy Summer Program (SASP) Mentor	2024
Lamat Program mentor	2021
Science Internship Program (SIP) mentor	2021
Mathematics Engineering Science Achievement (MESA) volunteer	2017-2019
Astronomy Club Leader	2017-2019

OBSERVING EXPERIENCE

Keck - Optical Spectroscopy (LRIS, 31 nights) / Adaptive Optics Imaging (1 night)
Shane (Lick) - Optical Spectroscopy (Kast, 22 nights)
SOAR - Optical Spectroscopy (Goodman, 7 nights)
KPNO Mayall - Optical Spectroscopy (KOSMOS, 5 nights)
Nickel (Lick) - Imaging (2 nights)
George H. Herbig Observational Astronomy Workshop. Lick Observatory, Mt. Hamilton, CA.
Shane and Nickel.

PRESENTATIONS

Talks

Transients in Middle Earth, Christchurch, New Zealand. *The Two Telescopes: SN Ia Redshift Evolution with Multiwavelength Spectroscopy*. February 2026.
AAS 247, Phoenix, Arizona. *SN 2025ogs: A Normal-Type Ia Supernova at $z = 2.05$* . January 2026.
One Hundred Years of Supernova Science, Stockholm, Sweden. *SN 2025ogs: A Normal-Type Ia Supernova at $z = 2.05$* . August 2025.
COSMOS Meeting 2025, Marseille, France. *The Spectroscopic Diversity of High- z Transients*. May 2025.
Transients from Space, Baltimore, MD. *Discovery of a Relativistic Stripped-envelope Type Ic-BL Supernova at $z = 2.83$ with JWST*. March 2025.
AAS 245, National Harbor, MD. *Discovery of a Relativistic Stripped-envelope Type Ic-BL Supernova at $z = 2.83$ with JWST*. January 2025.

COSMOS Meeting 2024, Tokyo, Japan. *Discovery of a Relativistic Stripped-envelope Type Ic-BL Supernova at $z = 2.83$ with JWST*. July 2024.

AAS 243, New Orleans, LA. *An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star*. January 2024.

SuperVirtual 2023. *Asymmetric Ejecta and Dust Emission in “Super-Chandrasekhar” Mass Type Ia Supernovae*. November 2023.

2023 MIAPbP Program “The Extragalactic Distance Scale and Cosmic Expansion in the Era of Large Surveys and the James Webb Telescope”, Garching, Germany. *An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star*. July 2023.

The Transient and Variable Universe Conference, UIUC. *An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star*. June 2023.

Time Domain Astronomy Seminar, Department of Astronomy, Caltech. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data*. November 2021.

Roman Science Team Community Briefing, Goddard Space Flight Center, NASA. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data*. November 2021.

Science Happy Hour, CIERA, Northwestern University. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data*. November 2021.

CAPS Seminar, Center for Astrophysical Surveys, University of Illinois Urbana-Champaign. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data*. October 2021.

Friday Scientific Lunch Talk, NOIRLab-Tucson, NOAO. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data*. October 2021.

Cosmology and Astronomy Seminar, Department of Physics and Astronomy, UC Davis. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data*. October 2021.

Astronomy Short Talk, Department of Astronomy, UC Berkeley. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data*. October 2021.

Conference - Progenitors of Type Ia Supernovae, Lijiang, China. *Using a Relational Database to Investigate Spectral Diversity in a Cosmological Sample*. August 2019.

FLASH Talk, Department of Astronomy, University of Arizona. *Investigating the Diversity of Type Ia Supernova Observations with a Relational Database*. August 2018.

FLASH Talk, Department of Astronomy and Astrophysics, UC Santa Cruz. *Familiar and Exotic Events in the Current Transient Landscape*. April 2018.

Posters

AAS 246. Anchorage, AK. “Recent Improvements to the Calibration of HST/STIS Spectroscopic Modes”. June 2025.

Accurate Flux Calibration in the Era of Space Astronomy and All-Surveys. Baltimore, MD. “How can we Further Improve Flux Calibration of the STIS Echelle Modes?”. October 2024.

First Year of Science with JWST Conference. Baltimore, MD. “*JWST reveals dust emission in a Super-Chandrasekhar-mass SN Ia*”. September 2023.

Keck Science Meeting. San Diego, CA. “*The Keck Type Ia Supernova Galaxy Survey*”. September 2021.

F.O.E. Fifty-One Erg Supernova Conference. Corvallis, OR. *A Relational Database for Type Ia Supernova Observations*. June 2017.

REFERENCES

Scientific

Ori Fox: ofox@stsci.edu (STScI, 410-338-6768)

Ryan Foley: foley@ucsc.edu (University of California Santa Cruz, 831-459-2835)

Technical

Joleen Carlberg: jcarlberg@stsci.edu (STScI, 667-218-6383)